Database Performance Tuning and Normalization

Problem Statement

Study and understand the concepts of Database Performance Tuning and Database Normalization.

Explain how performance tuning improves database efficiency and describe the different levels of normalization including:

- First Normal Form (1NF)
- Second Normal Form (2NF)
- Boyce-Codd Normal Form (BCNF)
- Fourth Normal Form (4NF)
- Fifth Normal Form (5NF)

Provide clear explanations with examples to demonstrate how normalization reduces redundancy and improves database design.

1. Performance Tuning

Definition

Database performance tuning is the process of optimizing database performance to ensure queries run efficiently and system resources are used effectively. The goal is to reduce response time and improve overall database performance.

Importance of Performance Tuning

- Improves query execution speed
- Reduces system resource usage
- Enhances scalability
- Improves user experience
- Ensures efficient data retrieval

Techniques Used in Performance Tuning

1. Indexing

Indexes help speed up data retrieval by allowing the database to locate records quickly

without scanning the entire table.

Example:

```
CREATE INDEX idx_employee_name

ON Employee(name);
```

2. Query Optimization

Writing efficient SQL queries helps reduce execution time.

Example:

Using specific columns instead of selecting all columns.

```
SELECT name, salary FROM Employee;
```

instead of

```
SELECT * FROM Employee;
```

3. Proper Database Design

Efficient schema design reduces redundancy and improves performance.

4. Using Joins Efficiently

Proper joins help retrieve related data efficiently.

5. Avoiding Unnecessary Data Retrieval

Fetching only required records improves performance.

Example:

```
SELECT * FROM Employee

WHERE salary > 50000;
```

2. Database Normalization

Definition

Normalization is a database design technique used to organize data in a database to reduce redundancy and improve data integrity. It divides large tables into smaller related tables and defines relationships between them.

First Normal Form (1NF)

Definition

A table is in First Normal Form if:

• All columns contain atomic (indivisible) values

• Each record is unique

• No repeating groups are allowed

Example

Not in 1NF

Student_ID Name

1

Subjects

Rahul Math, Science

2

Anita English, History

In 1NF

Student_ID Name Subject

1

Rahul Math

1

Rahul Science

2

Anita English

Anita History

Second Normal Form (2NF)

Definition

A table is in Second Normal Form if:

• It is already in 1NF

• All non-key attributes depend on the entire primary key

Example

Not in 2NF

Order_ID Product_ID Product_Name Quantity

Here Product_Name depends only on Product_ID, not on the full key.

In 2NF

Split into two tables:

Products Table

| Product_ID | Product_Name |

Orders Table

| Order_ID | Product_ID | Quantity |

Boyce-Codd Normal Form (BCNF)

Definition

A table is in BCNF if:

• It is already in 3NF

• Every determinant is a candidate key

Example

| Course | Instructor | Room |

If an instructor teaches only one course, then Instructor determines Course, causing

dependency issues.

Solution: Split into two tables.

Instructor Table

| Instructor | Course |

Course Room Table

| Course | Room |

Fourth Normal Form (4NF)

Definition

A table is in Fourth Normal Form if:

• It is already in BCNF

• It has no multi-valued dependencies

Example

| Student | Hobby | Language |

A student can have multiple hobbies and languages.

Solution

Split into two tables:

Student_Hobby

| Student | Hobby |

Student_Language

| Student | Language |

Fifth Normal Form (5NF)

Definition

A table is in Fifth Normal Form if:

• It is already in 4NF

• It cannot be further decomposed without losing information

• It removes join dependencies

Example

| Supplier | Product | Project |

This table may be decomposed into:

Supplier_Product

Supplier_Project

Product_Project

These tables can be joined back to reconstruct the original table.

Benefits of Normalization

• Eliminates data redundancy

• Improves data consistency

• Reduces storage space

• Improves database structure

• Simplifies database maintenance